

ANNEXURE V

Final Project Report 2023-27 Batch

1. Cover Page

- **Title:** Internship Report
- **Student Name:** Piyush Kumar Singh
- **Roll Number:** 150096723043
- **University / College Name:** ITM Skills University
- **Degree Program:** Bachelor of Technology (B.Tech) in Computer Science and Engineering
- **Internship Duration:** July 2025 – December 2025
- **Company / Organization Name:** OpenPaymnt Pvt. Ltd.
- **Submission Date:** 10 Jan 2026

2. Declaration

I hereby declare that this internship report titled “**Internship Report**” is submitted by me to **ITM Skills University** in partial fulfillment of the requirements for the degree of **Bachelor of Technology in Computer Science and Engineering**.

This report is my original work and has not been submitted earlier to any other university or institution. All the information included in this report is based on my actual internship work and learning experience.

Signature:



Name: Piyush Kumar Singh

Date: 10 Jan 2026

3. Acknowledgment

I would like to express my sincere gratitude to **ITM Skills University** for providing me the opportunity to undergo this internship as part of my academic curriculum. I am thankful to my **faculty guide** for their constant guidance and support throughout the internship period.

I would also like to thank **OpenPaymnt Pvt. Ltd.** for giving me the opportunity to work as an intern and gain real industry experience. I am especially grateful to my **mentors and senior team members** for their guidance, feedback, and continuous support, which helped me improve both technically and professionally.

4. Table of Contents

1. Cover Page
2. Declaration
3. Acknowledgment
4. Table of Contents
5. Introduction
6. Company Profile
7. Internship Role and Responsibilities

8. Work Description

8.1 UI/UX Design and Frontend Development Phase

8.2 AI, LLM, and RAG Development Phase

9. Learning Outcomes

10. Conclusion

11. References / Bibliography

5. Introduction

Purpose of Internship

The purpose of this internship was to gain practical industry experience and apply theoretical knowledge in real-world projects. It helped me understand professional workflows, teamwork, and modern technologies used in software development.

Objectives of the Internship

- To work on real projects in a professional environment.
- To improve technical skills in frontend development and AI technologies.
- To understand industry standards, tools, and best practices.
- To develop problem-solving and communication skills.

Relevance to Academic Curriculum

This internship strongly complemented my academic curriculum by providing hands-on experience in **web development, UI/UX design, AI systems, Large Language Models (LLMs), and Retrieval Augmented Generation (RAG)**, which are important areas in computer science today.

6. Company Profile

Company Name: OpenPaymnt Pvt. Ltd.

Location: London

Nature of Business:

OpenPaymnt Pvt. Ltd. is a technology-focused company working on modern digital solutions, including payment platforms, web applications, and AI-powered systems.

Technologies Used:

- Figma
- Javascript
- Tailwind CSS
- OpenAI, Anthropic, Gemini APIs
- Hugging Face
- LangChain
- Chroma Vector Database
- Gradio

Organization Structure:

The organization follows a team-based structure with developers, designers, AI engineers, and project managers working together under senior leadership.

7. Internship Role and Responsibilities

Internship Title: Software Development Intern

Department / Team: Web Development and AI Development Team

Responsibilities

- Designing UI layouts using Figma
 - Developing responsive landing pages using Next.js and Tailwind CSS
 - Fixing UI alignment and responsiveness issues
 - Working with LLM APIs and prompt engineering
 - Building AI assistants and RAG-based applications
 - Testing, debugging, and improving application performance
 - Participating in team reviews and feedback sessions.
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8. Work Description

8.1 UI/UX Design and Frontend Development Phase (July – Early October)

During the initial phase of my internship, I worked on **designing and developing a landing page** for OpenPaymnt.

Key Work Done:

- Studied modern startup landing pages for design inspiration
- Created wireframes and high-fidelity designs in Figma
- Designed responsive layouts for desktop and mobile
- Converted designs into frontend code using Next.js and Tailwind CSS
- Implemented hero section, navbar, features, team, contact, and footer sections
- Fixed alignment, spacing, and responsiveness issues
- Tested the website on different devices, screen sizes, and browsers
- Added hover effects, animations, and optimized images

Challenges Faced & Solutions:

- Mobile and tablet alignment issues were resolved using flexbox, grid, and proper spacing
 - Browser inconsistencies were fixed through testing and CSS adjustments
 - Performance issues were handled by image compression and layout optimization
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8.2 AI, LLM, and RAG Development Phase (Mid-October – December)

In the later phase, I worked on **AI systems, Large Language Models, and Retrieval Augmented Generation (RAG)**.

Key Work Done:

- Worked with OpenAI, Anthropic, and Gemini APIs
- Practiced prompt engineering, one-shot prompting, and streaming
- Built AI agents and multi-agent systems
- Created AI chat interfaces using Gradio
- Worked with Hugging Face open-source models
- Studied transformers, tokens, and context windows
- Built RAG pipelines using LangChain and Chroma
- Implemented document chunking, embeddings, and retrieval
- Evaluated RAG systems using retrieval and QA metrics
- Improved performance using reranking and preprocessing techniques

Challenges Faced & Solutions:

- Understanding RAG flow was challenging initially and solved through repeated practice
 - Optimizing chunk size and embeddings required experimentation and evaluation
 - Managing multiple models was handled by benchmarking and comparison
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9. Learning Outcomes

Technical Skills Gained

- UI/UX design using Figma
- Frontend development with Next.js and Tailwind CSS
- Responsive web design and optimization
- API integration and prompt engineering
- Working with LLMs and AI agents
- RAG development using LangChain and Chroma

Soft Skills Developed

- Communication and teamwork
- Time management
- Problem-solving
- Receiving and applying feedback

Industry Exposure

- Learned professional workflows
 - Understood real-world development cycles
 - Experienced code reviews and approvals
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10. Conclusion

- This internship at **OpenPaymnt Pvt. Ltd.** was a valuable learning experience. It helped me grow both technically and professionally. I gained hands-on experience in frontend development as well as advanced AI systems like LLMs and RAG.
- This experience has increased my confidence and prepared me for future roles in software development and AI-based applications. The skills learned during this internship will play a major role in shaping my career.